Build an AI-Powered customer 360 agent with Braze and Snowflake Cortex

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1. Overview

In this hands-on lab, you'll build an intelligent marketing analytics solution that combines Braze email engagement data with product reviews using Snowflake's Cortex AI capabilities. You'll create a natural language interface through Snowflake Intelligence that allows marketers to query both structured campaign performance data and unstructured customer reviews to receive AI-powered insights and recommendations.

Developed an **intelligent Customer 360 Marketing Analytics platform using Snowflake and Braze**, integrating email engagement data with customer product reviews. Data from multiple sources such as websites, mobile apps, CRM systems, transactions, and Braze campaign interactions was ingested and transformed in **Snowflake** to create a unified customer profile. Leveraged **Snowflake Cortex AI capabilities** to analyze both structured campaign performance data and unstructured customer reviews, generating AI-driven insights. Built a **natural language interface using Snowflake Intelligence** that enables marketers to query data easily and receive automated insights and recommendations. The platform supports **customer segmentation, behavioral analysis, campaign performance monitoring, and personalized engagement** across channels like email, push notifications, and in-app messaging, helping marketing teams improve targeting and campaign effectiveness.

1.1. What You'll Learn

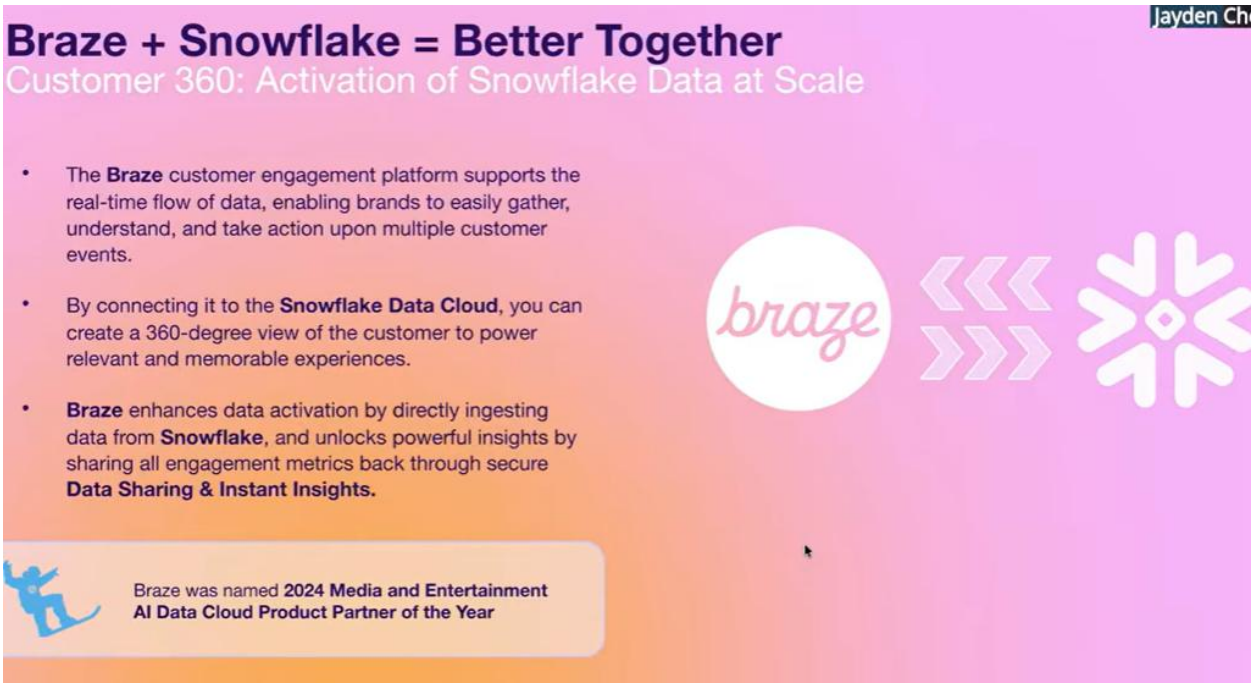
- How to set up Snowflake environment for Braze engagement data and product reviews
- How to create a Semantic View for structured data analysis with Cortex Analyst
- How to create a Cortex Search Service for unstructured product review analysis
- How to use Snowflake Intelligence to query both structured and unstructured data

1.2. What You'll Build

- A complete data pipeline for Braze email engagement data and product reviews
- A Semantic View that enables natural language queries on structured campaign data
- A Cortex Search Service for semantic search on product reviews (RAG)
- A Snowflake Intelligence agent that combines both tools for comprehensive marketing insights

1.3. Prerequisites

- Basic familiarity with SQL
- Access to Snowflake account with Cortex enabled
- Understanding of email marketing concepts



2. About Snowflake

Snowflake is a cloud-based data platform that enables scalable storage, processing, and analytics of structured, unstructured and semi-structured data with high flexibility and performance.

2.1. Snowflake key concepts and architecture

Snowflake is powered by an advanced data platform that is provided to you as a self-managed service. Snowflake's data platform brings together data storage, processing, and analytic solutions that are faster, easier to use, and far more flexible than traditional offerings.

Snowflake combines a completely new SQL query engine with an innovative architecture that is natively designed for the cloud. It offers full enterprise analytic database functionality, and unique features and capabilities.

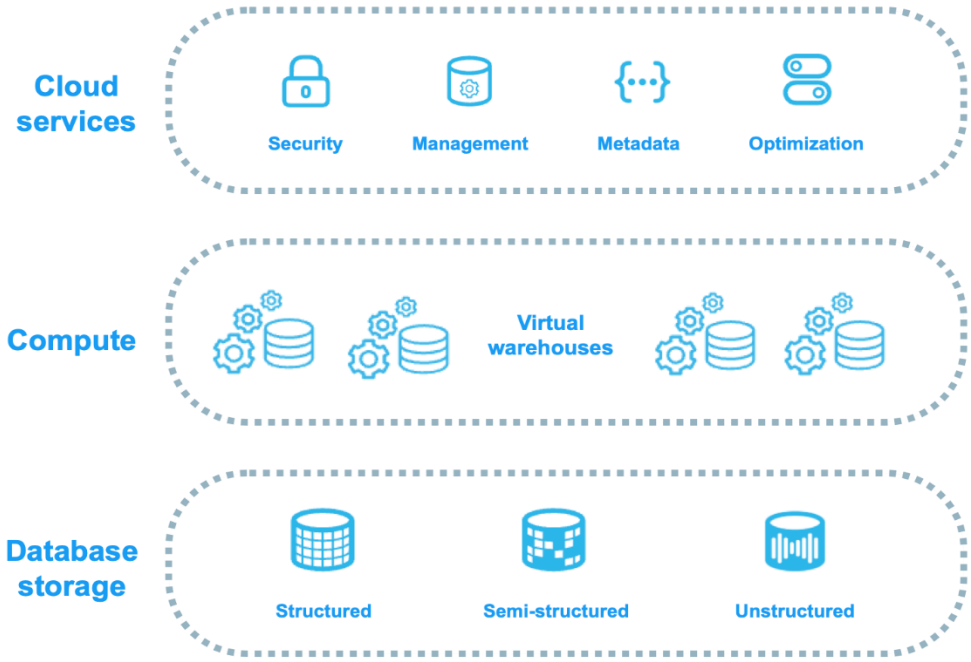
2.2. Data platform as a self-managed service

As a self-managed service, Snowflake has the following advantages:

- There is no hardware (virtual or physical) for you to select, install, configure, or manage.
- There is virtually no software for you to install, configure, or manage.
- Ongoing maintenance, management, upgrades, and tuning are handled by Snowflake.
- Snowflake uses public cloud infrastructure to host virtual compute instances and persistent data storage.
- Snowflake manages software updates and infrastructure so you don't have to.
- You can't install and run Snowflake locally or on private cloud infrastructures, whether on-premises or hosted.

2.3. Snowflake architecture

Snowflake's architecture is a hybrid of traditional shared-disk and shared-nothing database architectures. Similar to shared-disk architectures, Snowflake uses a central data repository for persisted data that is accessible from all compute nodes in the platform. But similar to shared-nothing architectures, Snowflake processes queries using massively parallel processing (MPP) compute clusters, where each node in the cluster stores a portion of the entire data set locally. This hybrid architecture, which is shown in the following diagram, offers the data management simplicity of a shared-disk architecture, but with the performance and scale-out benefits of a shared-nothing architecture:



Seamless Bi-Directional Native Integration

Enabled through Braze Cloud Data Ingestion and Snowflake Secure Data Sharing

- **Seamless Setup**: Connect Snowflake directly to Braze with just a few clicks.
- **Zero-Copy Data Access**: Access Braze data within Snowflake without copying or transferring information
- **Support for Customer and Non-Customer Data**: Sync customer data like attributes, events, purchases predictive data, segments, and more
- **Automated Syncing**: Sync Snowflake data to Braze as a one-off or on a recurring basis
- **Broad Data Integration**: Instantly join multiple, disparate data sets for comprehensive business insights (link to AWS).

2.4. Snowflake Data Cloud Overview

Snowflake is a **cloud-native data platform** designed to unify data storage, processing, and analytics without the complexity of traditional data warehouses.
It runs on major cloud providers (**AWS, Azure, Google Cloud**) and offers:

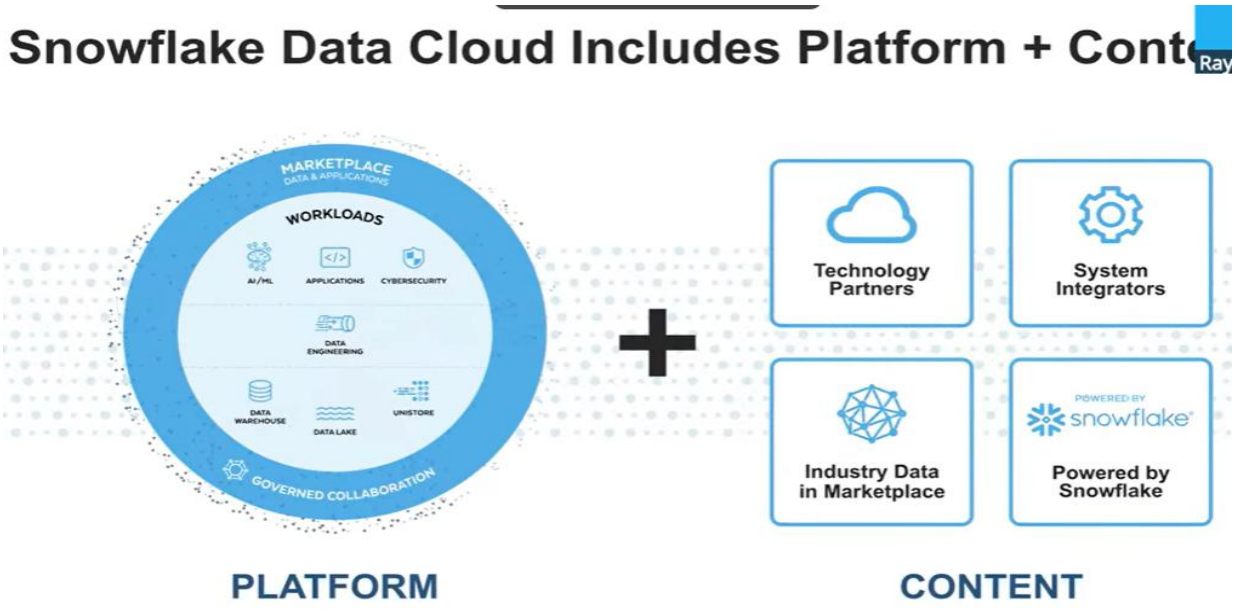
- **Multi-Cluster Shared Data Architecture**
  - Separates **compute** from **storage** for independent scaling.
  - Enables multiple workloads to run without contention.
- **Elastic Compute**
  - Auto-scaling and auto-suspend to optimize cost.
  - Supports both batch and real-time workloads.
- **Secure Data Sharing**
  - Share live, governed data across regions, clouds, and organizations without copying.
- **Cross-Cloud & Multi-Region**
  - Operates seamlessly across different cloud providers and geographies.
- **Built-in Governance**
  - Role-based access control, data masking, and compliance (HIPAA, GDPR, SOC 2, etc.).

- **Workload Support**
    - Data Warehousing
    - Data Lake
    - Data Engineering
    - Data Science & ML
    - Application Development
    - Unistore (transactional + analytical workloads)

2.5. Snowflake Content

The **Data Cloud** is not just infrastructure — it's also a **global ecosystem of data**:

- **Snowflake Marketplace**
    - Access to third-party datasets, APIs, and data services.
    - Categories: financial, healthcare, geospatial, marketing, etc.
- **Data Exchange**
    - Private or public exchanges for secure collaboration between partners.
- **Native App Framework**
    - Build and distribute data-driven applications directly in Snowflake.
- **Industry-Specific Data Clouds**
    - **Financial Services Data Cloud** – risk analysis, fraud detection.
    - **Healthcare & Life Sciences Data Cloud** – patient analytics, research.
    - **Retail Data Cloud** – supply chain optimization, personalization.
    - **Media Data Cloud** – audience analytics, ad performance.



3. About Braze

Braze is a Customer Engagement Platform used to power personalized, real-time marketing communications across every touchpoint, including email, push notifications, in-app messages, SMS, and WhatsApp. The platform captures a rich stream of first-party data, such as detailed user profiles, message engagement metrics (sends, opens, clicks), and custom behavioral events like purchases or conversions.

By centralizing this granular engagement data in Snowflake, you can join it with other business data to build a comprehensive 360-degree customer view. This enables deeper analytics on user behavior, more sophisticated customer segmentation, and the ability to power data-driven personalization.

Fundamentally, combining Braze and Snowflake allows you to fully understand the total business impact of your marketing workflows, campaigns and interactions across the ecosystem.

4. Architecture Overview

This project builds a **Customer 360 marketing analytics solution** using **Snowflake**, **Braze**, and **Snowflake Cortex**.

1. **Braze CSV Data** – Email campaign engagement data (sends, opens, clicks, unsubscribes) is exported from Braze as CSV files.
2. **Snowflake Stage** – The CSV files are uploaded into a Snowflake internal stage for secure storage before loading.
3. **Snowflake Tables** – Data from the stage is loaded into structured tables for analysis and transformation.
4. **Semantic View** – A semantic layer is created to define metrics, dimensions, and relationships, enabling natural language queries.
5. **Cortex Search** – Product reviews and customer feedback are indexed for semantic search using AI.
6. **Snowflake Intelligence Agent** – An AI agent combines structured campaign data and unstructured reviews to provide marketing insights through natural language questions.

This architecture enables marketers to **analyze campaign performance, understand customer sentiment, and optimize marketing strategies using AI-powered insights.**

5. Environment Setup

First, let's prepare your Snowflake environment and enable cross-region LLM usage.

5.1. Enable Cross-Region Cortex Access

Run the following command in a SQL worksheet to enable cross-region usage of LLMs, as your current region may be limited in which LLMs it can use:

```
ALTER ACCOUNT SET CORTEX_ENABLED_CROSS_REGION = 'ANY_REGION';
```

5.2. Create Database and Schema

Run the following SQL commands in a SQL worksheet to create the warehouse, database and schema:

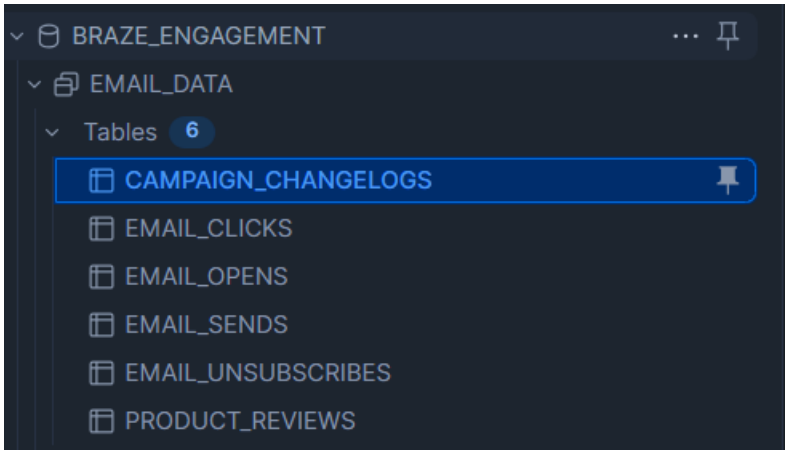
```
-- create the database
CREATE OR REPLACE DATABASE BRAZE_ENGAGEMENT;

-- create the schema
CREATE SCHEMA BRAZE_ENGAGEMENT.EMAIL_DATA;

-- switch to database and schema that was created
USE DATABASE BRAZE_ENGAGEMENT;
USE SCHEMA EMAIL_DATA;

-- create stage for raw data
CREATE OR REPLACE STAGE EMAIL_STAGE DIRECTORY = (ENABLE = TRUE);
```

6. Create Data Tables

- Now we'll create tables for your email engagement data, campaign changelog data, and product reviews. The full table schemas for Braze engagement data can be found here.

- You will be prompted to download the files shortly from GDRIVE or an AWS bucket.
- **Positive** : **Note on Data Types**: For this lab, we're using `TIMESTAMP_LTZ` datatype rather than a Number representing a UNIX timestamp for easier use in setting up our semantic view.
- Run the following SQL to create all necessary tables:

After running this SQL, navigate to **Data** > **Databases** and you should see your BRAZE_ENGAGEMENT database, EMAIL_DATA schema, and the 6 tables you just created.



7. Upload Sample Data

Now we'll upload sample CSV files to populate our tables with demo data.

7.1. Download Sample Data

Download the following CSV files (also available through the AWS bucket provided):

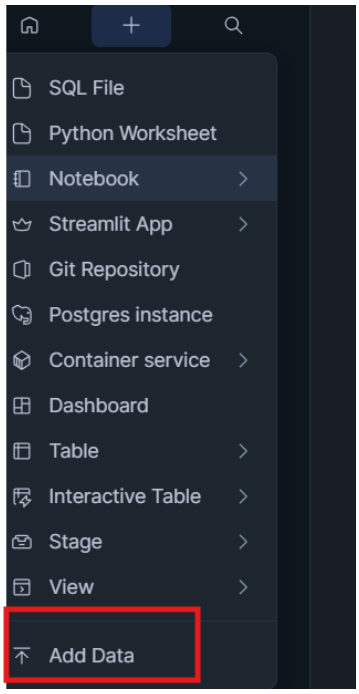
- USERS_MESSAGES_EMAIL_UNSUBSCRIBE_VIEW.csv
- USERS_MESSAGES_EMAIL_CLICK_VIEW.csv
- USERS_MESSAGES_EMAIL_OPEN_VIEW.csv
- USERS_MESSAGES_EMAIL_SEND_VIEW.csv
- CHANGELOGS_CAMPAIGN_VIEW.csv

Positive : **Note**: Product reviews data will be inserted directly via SQL in the next section, so no CSV upload is required for that table.

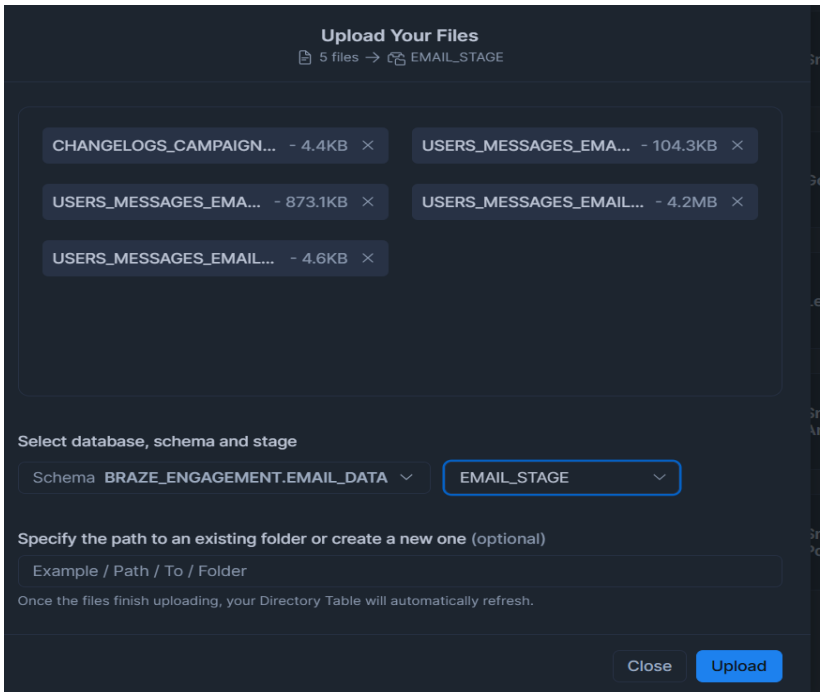
7.2. Upload Files to stage

To upload the data files:

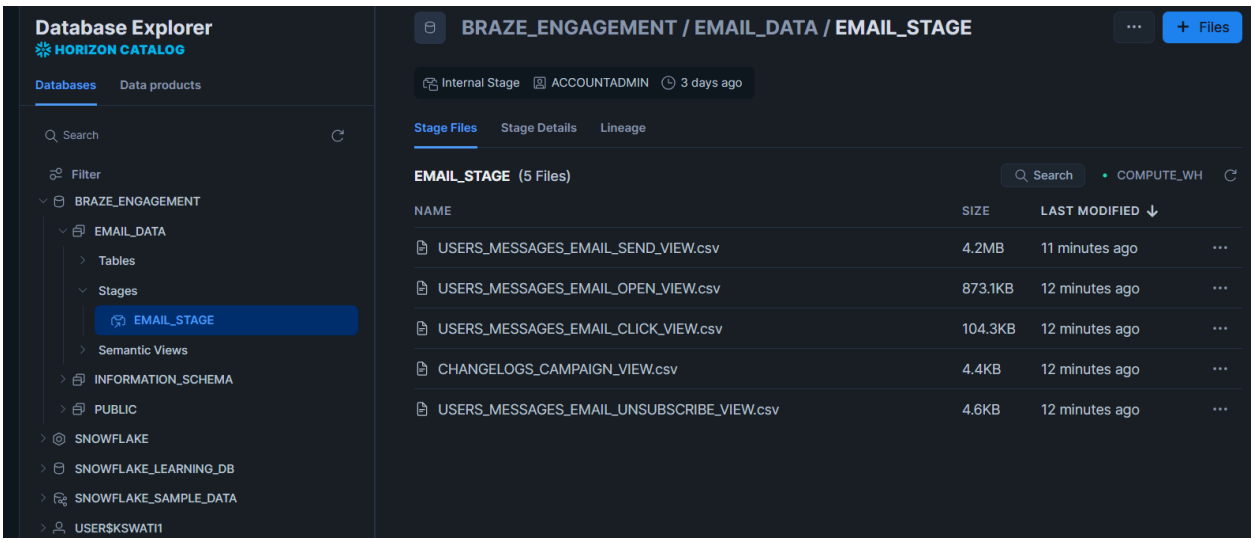
1. Choose **Create** in the left-hand navigation and select **Add Data** from the dropdown



2. On the Add Data page, select **Load files into a stage**
3. Select the five Braze CSV files that you want to upload (listed above)
4. Select BRAZE_ENGAGEMENT as Database, EMAIL_DATA as Schema, and EMAIL_STAGE as Stage
5. Click **Upload**



Navigate to **Data > Databases**, click into your BRAZE_ENGAGEMENT Database, EMAIL_DATA Schema, and the EMAIL_STAGE. You should see your 5 files listed.



8. Load data into table

Now we'll load the data from your CSV files into their respective tables. By two ways

1) By copy command
2) Direct load from stage

Verify Data Load

```
149 SELECT * FROM BRAZE_ENGAGEMENT.EMAIL_DATA.EMAIL_UNSUBSCRIBES LIMIT 10;
```

Results (Mar 8, 5:25:19 PM) | Results (just now)

Table | Chart | 10 rows | 271ms

	REVIEW_ID	USER_ID	RATING	REVIEW_TEXT	PRODUCT_
	R-10... 10.0% R-10... 10.0% +8 more	06Y1ResA... 10.0% 09cW2jhml... 10.0% +8 more	1 5	Fantastic graphic novel! Gripping story and bea... 10.0% Gorgeous leather belt. High quality leather and ... 10.0% +8 more	Positive Negative +1 more
1	R-1001	06Y1ResADlZutwm	5	The denim jeans fit perfectly and the quality is exceptional f	Positive
2	R-1002	0MJWsStTQuYWYq7	1	My coffee maker stopped working after the first week. Cust	Negative
3	R-1003	1NQHFvWpLpo5LkC	4	Love these wireless earbuds! Great sound for the price, thou	Positive
4	R-1004	1N6R4EtkRuKEjFP	3	The travel guide was adequate, but definitely focused on to	Neutral
5	R-1005	1OEy7FFzL9n5FTl	5	Gorgeous leather belt. High quality leather and the buckle lo	Positive
6	R-1006	09cW2jhmlPQ6CZ8	2	The Desk Organizer was broken upon arrival. The plastic is	Negative
7	R-1007	0PCkfnjde7Pz7Ya	5	This portable charger is a lifesaver. Holds multiple charges a	Positive

9. Create Semantic View

Now we'll create a Semantic View that enables natural language queries on our structured email engagement data using Cortex Analyst. Unlike a traditional semantic model YAML file, a Semantic View is a first-class database object that can be managed with SQL.

The semantic view defines:

- **5 logical tables** with primary keys and relationships
- **Dimensions** for time, timezone, gender, user IDs, and campaign information
- **8 metrics** for total and unique counts of sends, opens, clicks, and unsubscribes
- **AI instructions** to guide Cortex Analyst in generating accurate SQL

Verify the semantic view was created

Cortex Analyst | Help | ACCOUNTADMIN • COMPUTE_WH (X-Small) | Create new

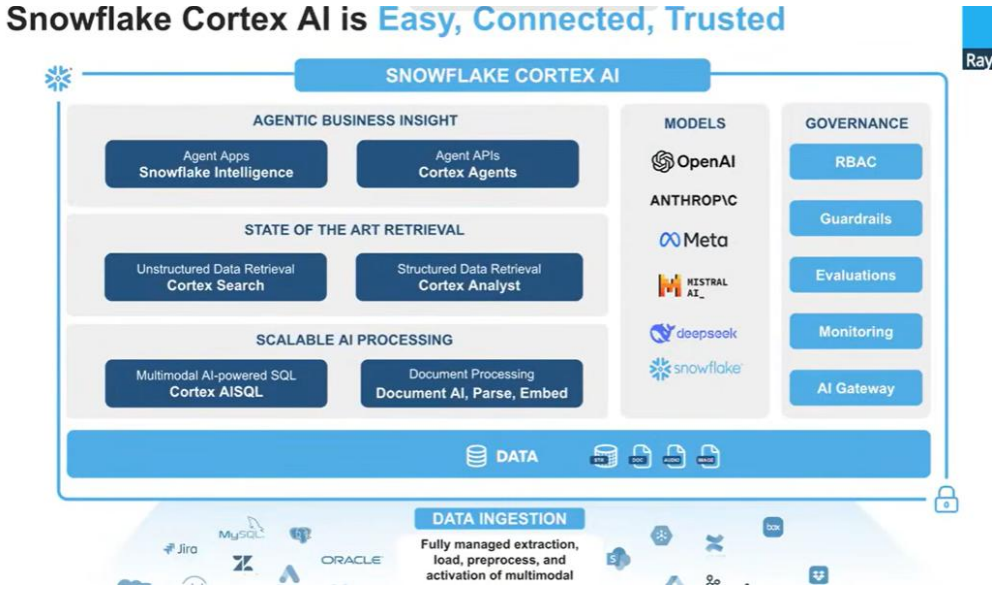
Semantic views | Semantic models

BRAZE_ENGAGEMENT.EMAIL_DATA | Search

NAME	DESCRIPTION	CREATED
CAMPAIGN_ANALYTICS_VIEW	Semantic view for Braze email engagement analytics i...	3 days, 19 hours ago

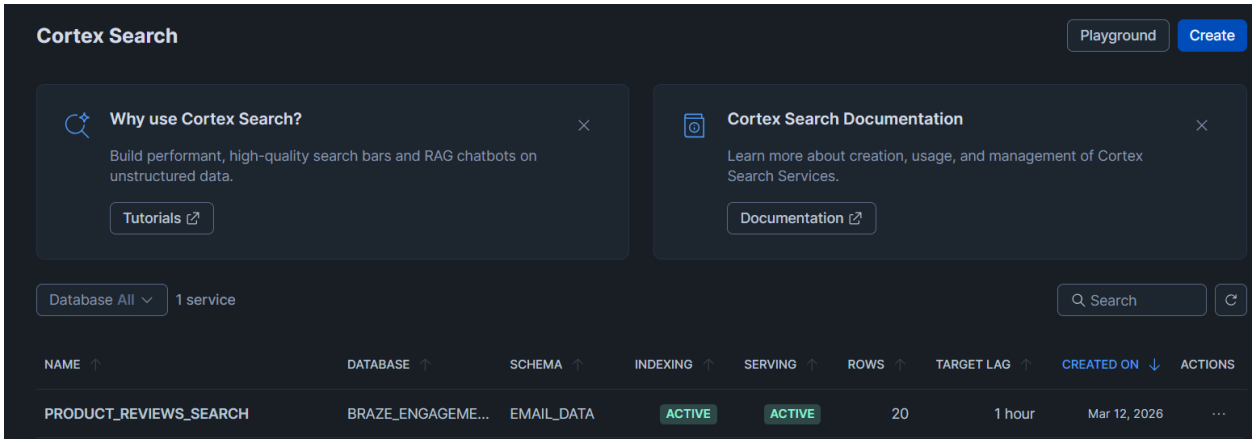
10. Create Cortex Search service

Now we'll create a Cortex Search Service to enable semantic search on our product reviews. This allows us to use RAG (Retrieval Augmented Generation) to analyze unstructured customer feedback.



The Cortex Search Service:

- Indexes the REVIEW_TEXT column for semantic search
- Includes filterable attributes like rating, sentiment, location, category, and item name
- Automatically updates with a 1-hour target lag
- Enables natural language queries over unstructured review data

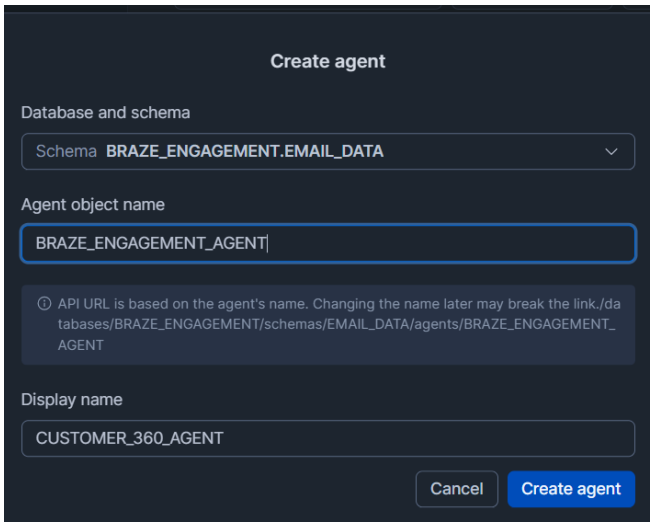


11. Configure Snowflake Intelligence

Now we'll set up Snowflake Intelligence with an agent that combines both the Semantic View (for structured campaign data) and the Cortex Search Service (for unstructured product reviews).

11.1. Access Snowflake intelligence

1) In Snowsight, navigate to **AI & ML** > **Snowflake Intelligence**
2) Click **+ New Agent** to create a new agent



11.2. Configure the Agent

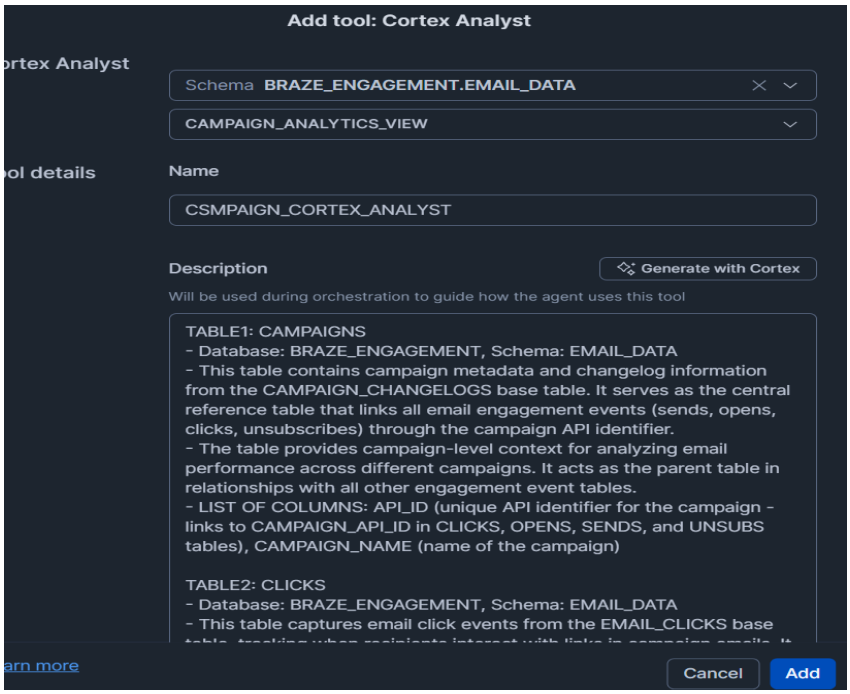
- Set up your agent with the following configuration:
- **Agent Name:** CUSTOMER_360_AGENT
- **Description:** An AI assistant that helps marketers analyze email campaign performance and customer product reviews to optimize marketing strategies.
- **Sample Questions:**
  i. How many email was sent?
  ii. What are the most common positive and negative themes in customer product reviews, and how do they relate to recent email campaigns?
  iii. Which products mentioned in negative reviews were promoted in recent email campaigns, and how did those campaigns perform?

11.3. Add tool

Add the following tools to your agent:

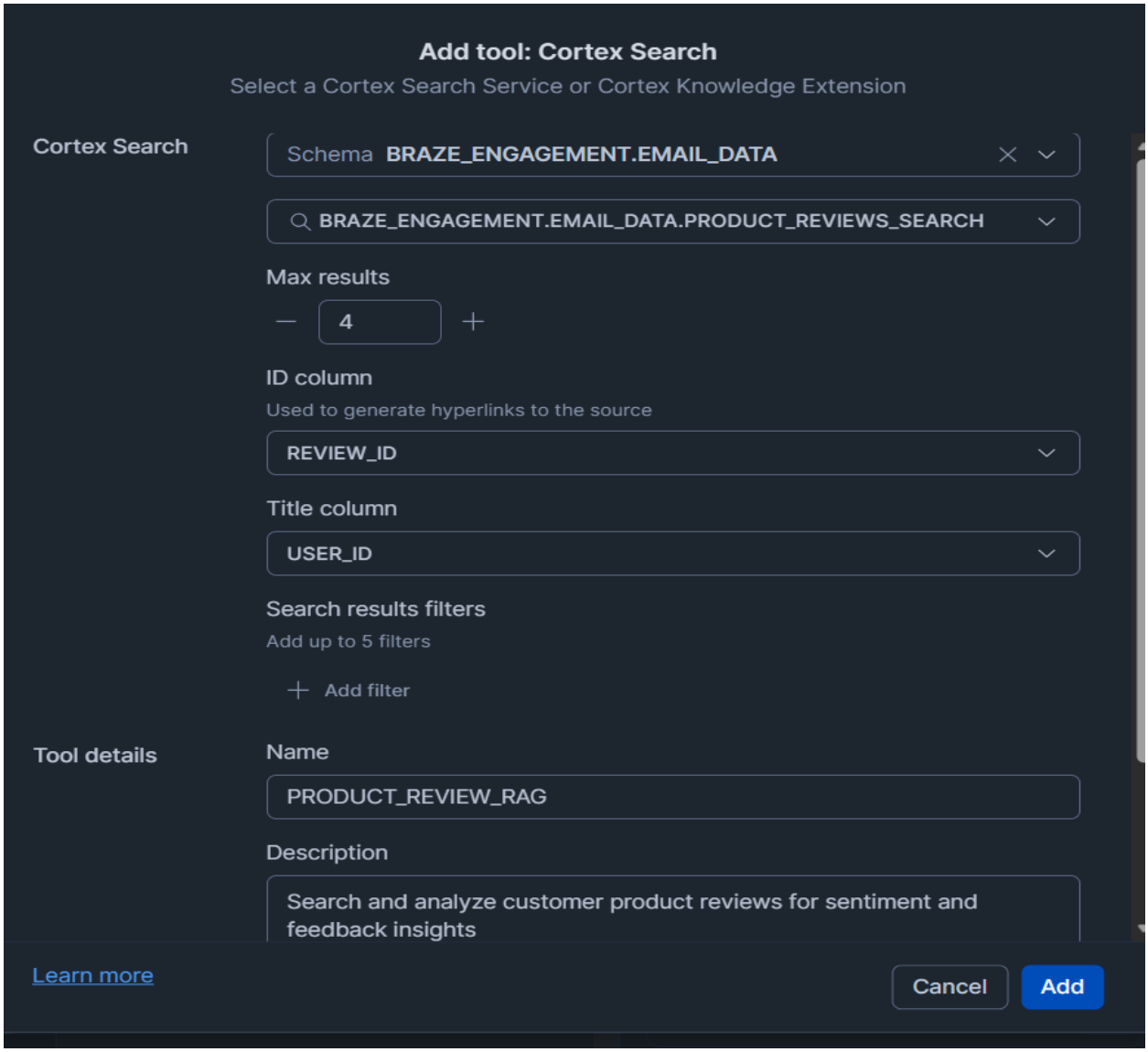
Tool 1: Semantic View (Structured Data)

- Type: Cortex Analyst
- Semantic View: BRAZE_ENGAGEMENT.EMAIL_DATA.CAMPAIGN_ANALYTICS_VIEW
- Description: Query email campaign engagement data including sends, opens, clicks, and unsubscribes



Tool 2: Cortex Search (Unstructured Data)

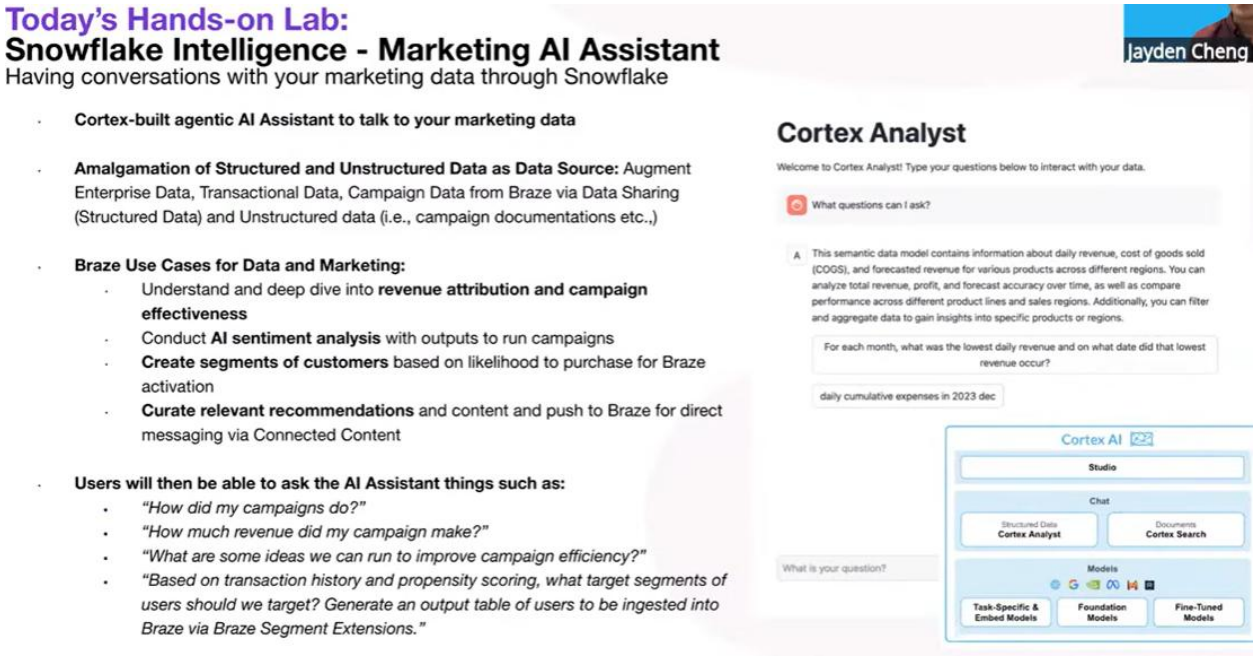
- Type: Cortex Search
- Search Service: BRAZE_ENGAGEMENT.EMAIL_DATA.PRODUCT_REVIEWS_SEARCH
- Description: Search and analyze customer product reviews for sentiment and feedback insights.
- Select ID column and title column as per image.



11.4. Save and activate

Click **Save** to create your agent. The agent is now ready to answer questions using both structured campaign data and unstructured product reviews.

12. Using Snowflake Intelligence



Now let's explore how to use Snowflake Intelligence to gain marketing insights from both your structured and unstructured data.

a) Sample Questions for Structured Data (Campaign Analytics)

Try these questions to analyze your email campaign performance:

Basic Metrics:

1. "How many emails were sent in total?"
2. "What is the overall open rate across all campaigns?"
3. "Which campaign had the most clicks?"

Time-based Analysis:

1. "Show me email engagement trends by month in 2024"
2. "Which campaigns had the highest unsubscribe rates in the first half of 2025?"

Segmentation:

1. "What timezone had the highest engagement for our campaigns?"
2. "Break down opens and clicks by gender"

Campaign Performance:

1. "What campaign had the most unsubscribes over all time?"

2. "Compare engagement metrics across all campaigns"

b) Sample Questions for Unstructured Data (Product Reviews)

Try these questions to analyze customer feedback:

Sentiment Analysis:

1. "What are customers saying about electronics products?"
2. "What are the main complaints in negative reviews?"
3. "Summarize the positive feedback we're receiving"

Product Insights:

1. "Which products have the best customer feedback?"
2. "What issues are customers having with apparel?"
3. "What do customers think about our home goods?"

Channel Analysis:

1. "How do reviews differ between online and retail store purchases?"
2. "What are app customers saying about their purchases?"

c) Sample Questions Combining Both Data Sources

These questions demonstrate the power of combining structured and unstructured data:

Strategic Planning:

1. "What should my email marketing optimization strategy be based on campaign performance and customer feedback?"
2. "Based on product reviews and email engagement, which product categories should we focus our next campaign on?"
3. "What improvements should we make to our marketing approach based on both campaign metrics and customer sentiment?"

Customer Experience:

1. "How can we reduce unsubscribes based on what customers are saying in reviews?"
2. "Which products with positive reviews should we highlight in our email campaigns?"
3. "What messaging resonates with customers based on review feedback and click patterns?"

d) Understanding the Output

When you ask questions, Snowflake Intelligence will:

1. **Determine the appropriate tool(s)** to use based on your question
2. **For structured data questions**: Generate and execute SQL queries against the semantic view
3. **For unstructured data questions**: Perform semantic search on product reviews and synthesize insights
4. **For combined questions**: Use both tools to provide comprehensive answers
5. **Provide actionable insights** tailored for marketing decision-making

13. Conclusion

Congratulations! You've successfully built an intelligent marketing analytics solution that combines Braze email engagement data with product reviews using Snowflake's Cortex AI capabilities.

What You Learned

You now have experience with:

- Setting up Snowflake environment for Braze engagement data and product reviews
- Creating Semantic Views for structured data analysis with SQL DDL
- Creating Cortex Search Services for unstructured data analysis (RAG)
- Using Snowflake Intelligence to query both structured and unstructured data
- Combining multiple AI tools for comprehensive marketing insights

What You Built

Your complete solution includes:

- **Data Pipeline**: Stores and processes Braze email engagement data and product reviews in Snowflake
- **Semantic View**: Enables natural language queries on structured campaign data through Cortex Analyst
- **Cortex Search Service**: Enables semantic search on unstructured product reviews
- **Snowflake Intelligence Agent**: Combines both tools for comprehensive marketing analysis

Key Capabilities

Your solution can:

- Query email campaign performance using natural language

- Search and analyze customer product reviews semantically
- Combine insights from both structured and unstructured data
- Provide actionable recommendations for marketing optimization
- Maintain data integrity by grounding responses in actual data

Next Steps

To extend this solution, consider:

- Adding more Braze data tables (push notifications, in-app messages, etc.)
- Incorporating additional product review sources
- Creating scheduled data refreshes for real-time insights
- Building custom agents for specific marketing use cases
- Implementing user access controls and sharing capabilities

14. Recourses

1) Swatikhedekar/Build-AI-Powered-Campaign-Analytics-with-Braze-and-Snowflake-Cortex
2) **AI-Powered Campaign Analytics with Braze and Snowflake Cortex**
3) Practitioner
4) https://try.braze.com/free_trial
5) Overview of semantic views | Snowflake Documentation
6) Export Braze Data